

Proposal #25

Remote Eye-Tracking *Software Requirement Specifications*

CS Capstone Team 12



PFW CS Department -Sponsor

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Functional Requirements

Feature List

The system must provide the following features:

- Users shall be able to create, edit, and delete their profiles, with the exception of "Guests", whose data will not be saved.
- Users shall have the capability to create new projects, including the addition of tasks, descriptions, and URLs.
- Users shall be able to share projects with other users or guests for analysis.
- The system shall perform real-time tracking of user eye movements using standard laptop webcams.
- The system shall record user sessions, including screen recordings with an overlay of eye-tracking data.
- Users shall have the option to log in, register, and select various roles, such as Analyst, Project Manager, and Admin.
- A user interface shall be provided for project creation, analysis, and viewing recorded sessions alongside data.

Data Handling

The system shall ensure secure and efficient data handling, including:

- Validation of input data for accuracy and completeness.
- Secure storage of user data, session recordings, and analysis results in the database.
- Encryption of all stored and transmitted data to protect user privacy.
- Data retrieval functionalities that support filtering and searching for analysis purposes based on the project.

System Actions

The system shall perform specific actions in response to user inputs or events:

- Generation of reports on user behavior based on eye-tracking data, eg) time taken to complete task.
- Requesting users to agree to our TOS and give certain browser based permissions.
- Notification of users regarding session status and progress.
- Display of real-time eye movement on the screen as a visual overlay.
- Authentication of users with access controls based on roles.

Integration Requirements

The system must integrate with external APIs and systems, including:

- Integration with third-party APIs and libraries, such as Webgazer.js, to enable eye-tracking functionality.
- Connection to external databases for data storage and retrieval if necessary.

User Roles and Permissions

The system shall define distinct roles and permissions for users:

- Admins shall manage all user accounts underneath them, projects, and system settings.
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- Project Managers shall create projects, share them, and analyze results.
- Analysts shall have access to assigned projects and their analysis data.
- Guests shall have limited access, restricted to viewing shared projects without editing or creating capabilities.

Non-Functional Requirements

Performance Requirements

The system shall meet the following performance standards:

- Pages must load within 3 seconds under normal usage conditions.
- The system shall support real-time eye-tracking and screen recording without noticeable latency.

Security Requirements

The system must adhere to strict security protocols:

- AES-256 encryption shall be used for all data in transit and at rest.
- Multi-factor authentication shall be required for user login.

Usability Requirements

The system shall be user-friendly and accessible:

- Compliance with WCAG 2.1 Level AA accessibility guidelines is mandatory.
- The user interface shall be intuitive, with tooltips and help documentation readily available.

Reliability and Availability

The system shall ensure high reliability and availability:

- The system must maintain a high percentage uptime.
- Data backups shall occur every 24 hours to prevent loss.

Maintainability Requirements

The system shall be easy to maintain and update:

- The codebase shall be modular to facilitate updates and bug fixes.
- Version control shall be employed to manage changes and support collaboration.

Scalability Requirements

The system must support growth in users, data, and transactions:

- The system shall support concurrent users without performance degradation.
- The system shall efficiently handle large datasets, including eye-tracking data and video recordings.

Use Cases/User Stories

User Scenarios

Various scenarios describe how different users will interact with the system:

- As a registered user, I want to reset my password to regain access to my account.
 - As a Project Manager, I want to create a new project and assign tasks to team members.
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- As a Guest, I want to view shared projects to understand user behavior without needing an account.

Interaction Flows

The system shall provide detailed interaction flows for each user scenario:

- "Forgot Password" process: The user selects "Forgot Password," receives a reset link via email, and sets a new password.
- Admin management flow: Admin navigates to the user management panel to add or remove users.
- Project sharing flow: The Project Manager shares a project link, and a Guest user accesses the link, views the project, and starts analysis.

User Roles and Goals

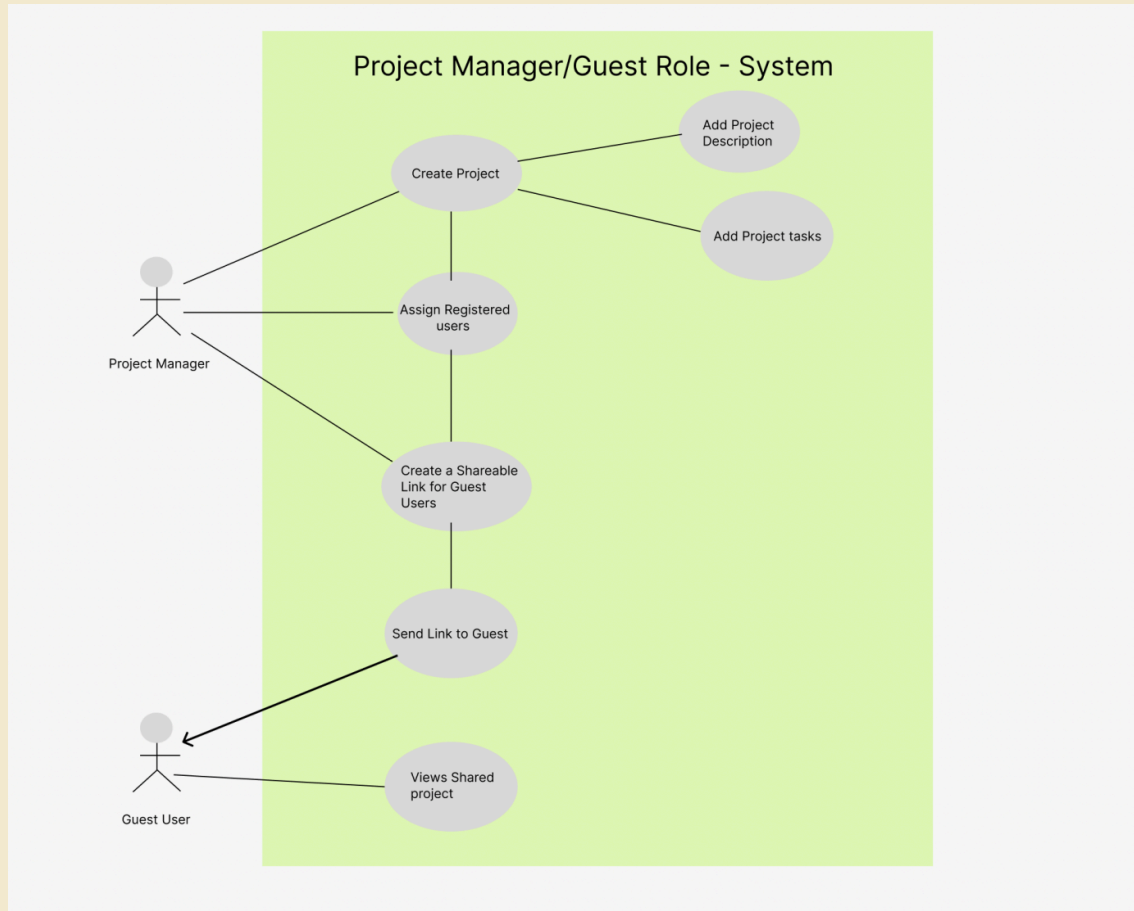
The system shall support the following user roles and their objectives:

- Admin users manage system-wide settings, user accounts, and data integrity.
- Project Managers create, manage, and analyze UX testing projects.
- Student users engage with assigned projects, perform testing, and review their analysis.
- Guest users access and view shared projects for feedback purposes.

Success and Failure Conditions

The system shall define conditions for successful and failed interactions:

- Success: The user successfully resets their password and logs in.
 - Failure: The user receives an error message for an invalid reset link.
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User Interface Requirements

Design Principles

The user interface shall adhere to the following design standards:

- A consistent color scheme and typography shall be used throughout the interface.
- The design shall prioritize simplicity and ease of navigation.

Wireframes

The system shall include wireframes or mockups to represent the user interface visually:

- The login page shall include fields for username and password, a "Login" button, and a "Forgot Password" link.
- The project creation page shall provide fields for the project title, description, tasks, and sharing options.

Navigation Requirements

The system shall provide clear navigation paths:

- The main navigation menu shall be accessible from all pages and include links to Home, Profile, Settings, and Logout.
- Breadcrumb navigation shall be implemented to enhance usability.

Accessibility Features

The system shall ensure accessibility for all users:

- All interactive elements must be keyboard accessible, and alternative text shall be provided for all images.

Sprints

Sprint 1

Goals: Setup Flask Server

Accomplished: Established Flask server with route conventions and user authentication.

Comments: For our first sprint, we decided to set up the backbone of our design.

We know that one of our project goals is to integrate our application into a Proxmox server later on, but we wanted to get started with something before we integrate with it.

We first wanted to integrate a Flask server to get us started and allow us to begin development before integrating with our sponsor's server.

For our first sprint, we were able to do just that.

We got Flask up and running and began working on the structure of our design.

In order to test the Flask server functionality, we set up our login and register pages.

We also made sure to organize it so we could begin working on the different web pages necessary for the application.

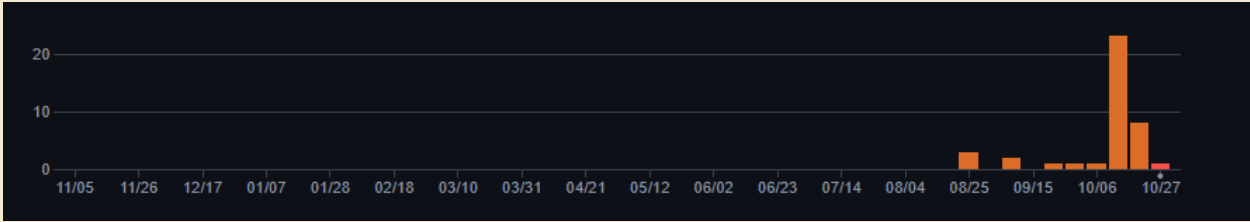
Sprint 2

Goals: Establish Individual Pages

Accomplished: Began Implementing.

Comments: In our second sprint, we decided to divide and conquer. We all wanted a part in the development as full stack developers, and this is the point we get to do just that. We split up the different pages necessary for the application amongst the group in an effort to get hands-on experience in front-end and back-end development. We chose to work on the Create, View and Edit portion of the application. We strategically chose these pages so we would have the chance to work together, but also separately engage in front-end and back-end development.

During this process, we were each able to get set up and get started. We are still in the process of developing these pages and how they interact with the flask server. During this sprint, we were all able to access GitHub and get our project setup for all of us to interact with. There were some minor issues learning how to navigate the project and get everything established to do our work. In the end, we were able to get all these concerns addressed and we have each begun working on these pages. We plan on completing this sprint by Sunday.



@Capstone#25

Backlog

Team capacity

Current iteration

Roadmap

My items

New view

Filter by keyword or by field

Todo

18/58

Estimate

This item hasn't been started

Draft

Create project page front end -Kyle

Draft

Create project page backend -Kyle

Draft

View Project back end -Sulaiman

Draft

Edit Project back end -Logan

Draft

Admin Control Panel front end -Sulaiman

Draft

Admin Control Panel Back end -Sulaiman

Draft

Settings Page Back end -Kyle

Draft

Calibration Test Front end -Kyle

Draft

Dashboard (home) backend - Dominic

Draft

Calibration Test back end -Kyle

In Progress

16/58

Estimate

This is activity being worked on

Draft

Settings Page Front end -Kyle

Draft

Dashboard (home) front end -Dominic

Draft

View Project page front end -Sulaiman

Draft

Edit Project front end -Logan

Semi Done

0/

Estimate

"Done" until it needs to be revisited

Draft

Register backend - Dominic

Draft

Register front end -Dominic

Done

9/188

Estimate

This has been completed

Draft

Initial Flask Server

Draft

Login front end -Dominic

Draft

Login backend -Dominic

Stretch Goals

2/20

Estimate

Draft

Mobile Support

Draft

More Auto Data Analytics